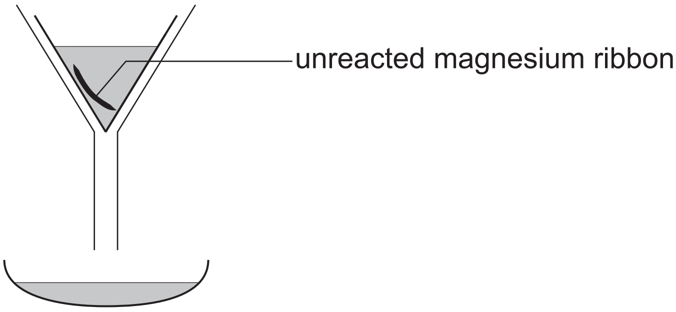
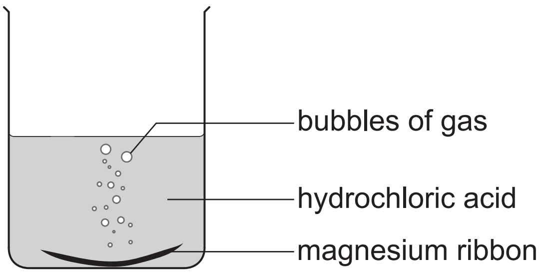
Answer **all** questions.

1. (a) A student carried out an experiment to prepare magnesium chloride crystals.

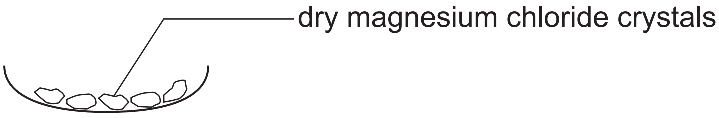
Diagrams **A**, **B** and **C** show the stages of the experiment she carried out. The stages are **not** in the correct order.



A



B



C

Examiner only

(i) I. Give the **letter** that shows the **first** stage of the experiment. [1]

Letter ....

II. Give the **letter** of the stage that shows filtration. [1]

Letter ....

(ii) The gas formed in the reaction pops with a lighted splint.

Underline the name of this gas. [1]

oxygen **hydrogen** **carbon dioxide**

(iii) Magnesium chloride contains Mg^{2+} and Cl^{-} ions.

Tick (✓) the box next to the formula of magnesium chloride. [1]

$MgCl_2$ ☐ **Mg_2Cl** ☐ **MgCl** ☐ **Mg_2Cl_2** ☐

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(b) The word equations below show reactions of acids.

Underline the substance in each box that correctly completes the equation. [2]

zinc oxide + [**hydrochloric acid** / **nitric acid** / **sulfuric acid**] ⟶ zinc sulfate + water

sodium hydroxide + nitric acid ⟶ [**sodium chloride** / **sodium nitrate** / **sodium sulfate**] + water

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